

Optimizing the Distributing of Donor Hearts

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Importance of the Heart

Short Term

- General Life

Long Term

- Vascular
- Anaerobic
- Immune System

Heart Transplant Statistics

**Current
Waitlist**

5,896

**New
Additions**

4,711

**Transplants
Performed**

4,565

Dataset

Clinical trial dataset containing:

- Drug Status (D-penicillamine vs Placebo)
- Patient Data (Sex and Age)
- Medical Data (Cholesterol, Platelets, SGOL, etc)
- Disease Severity (Stage)
- Patient's Outcome (Alive/Dead and Duration)

Important Features:

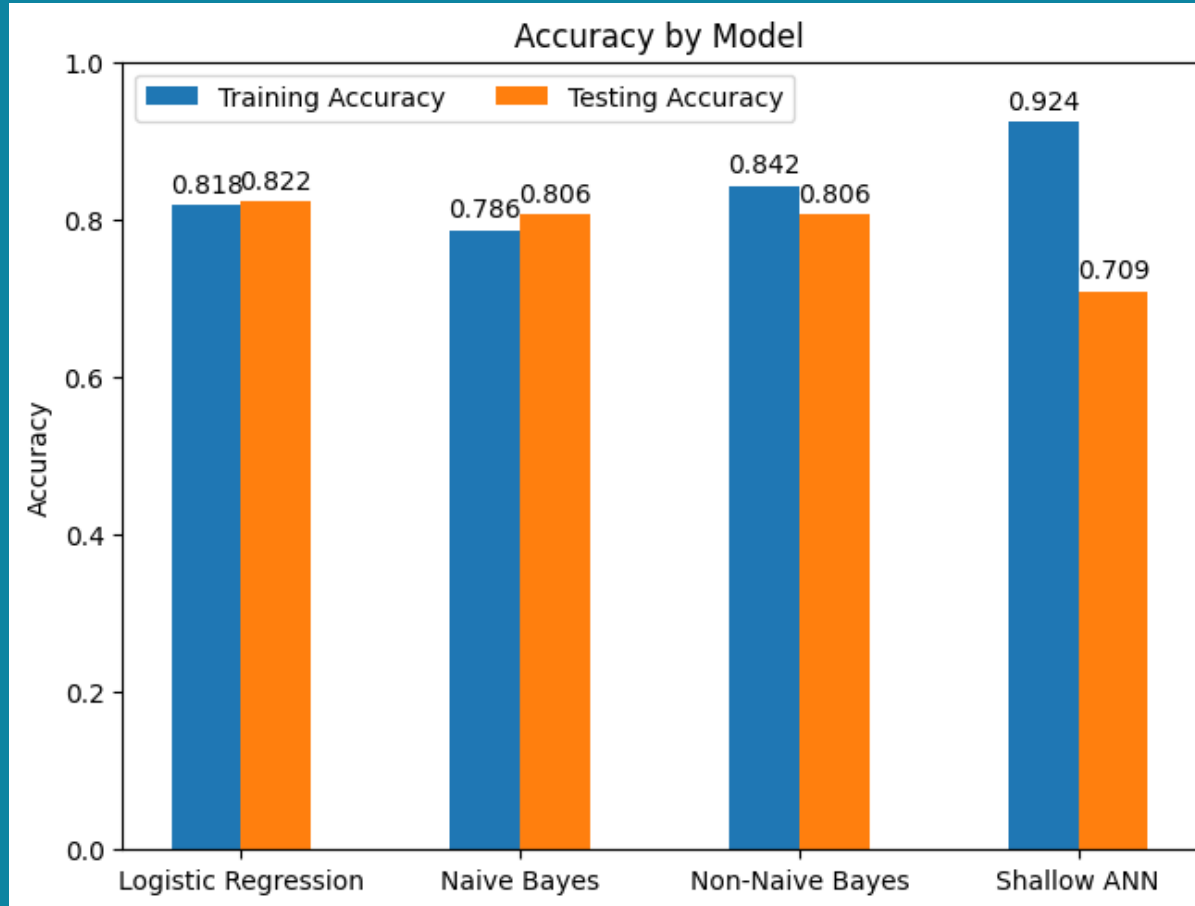
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Dataset Problems

Problems with a Clinical Trial Dataset:

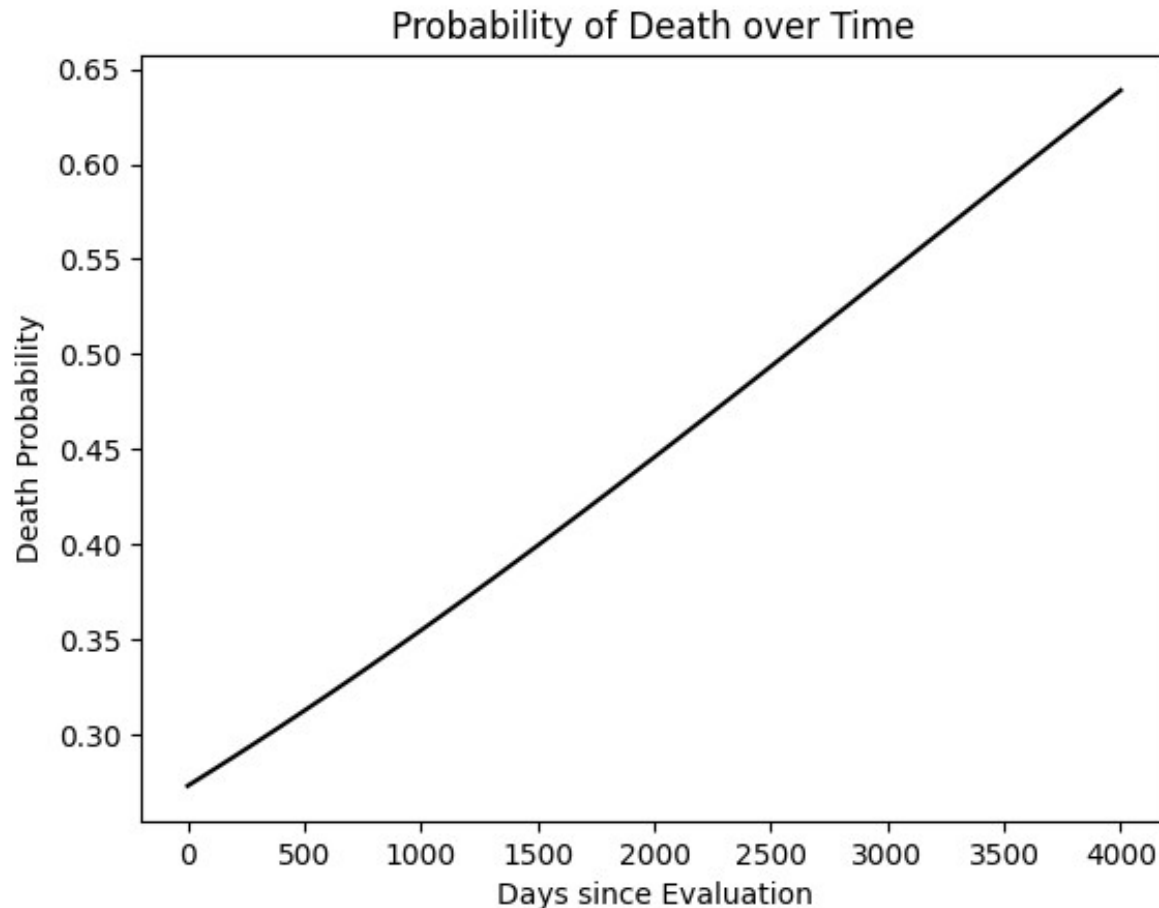
- Small Amount of Data
- Different Goal
- Biased Focus Group
- Single Check-up
- Limited Duration
- Mixed Ending Status

Model Accuracy



- Shallow ANN performs the best on Training Data
- Logistic Regression performs best on Testing Data

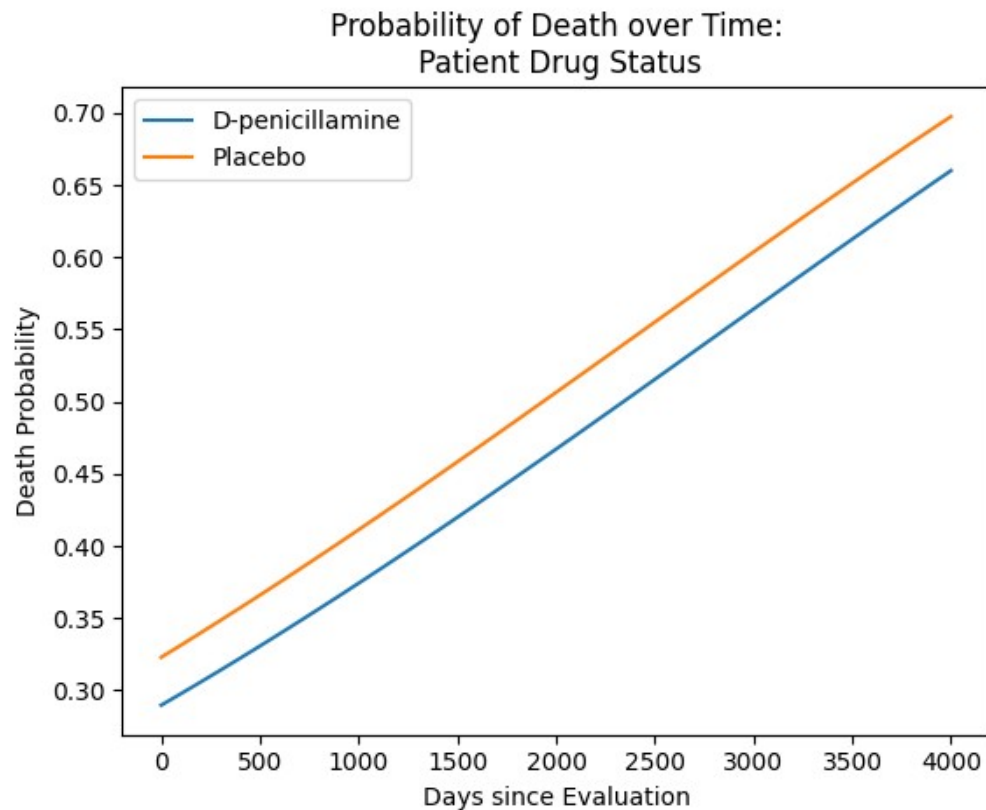
Life Expectancy



- **Linear increase over time**
- **Starts at around 25% at evaluation**
- **Maximum of 65% after 11 years**

Life Expectancy

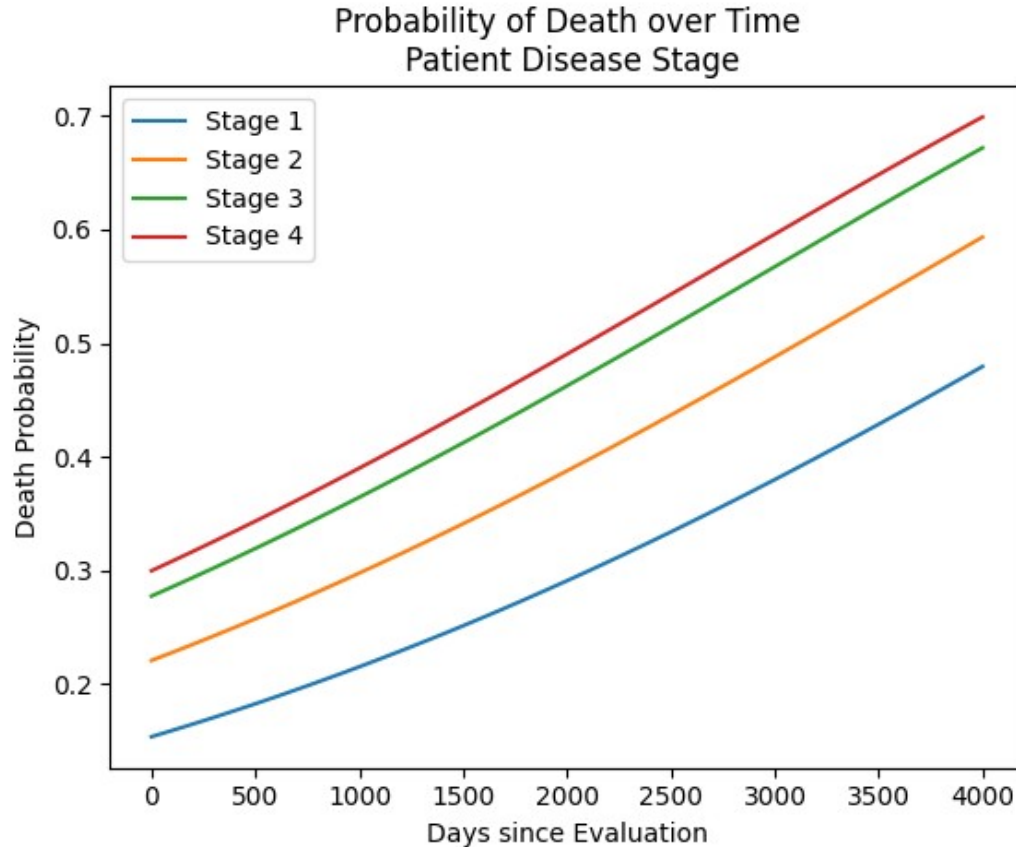
Drug



- **D-penicillamine provides a small benefit of around 3%**
- **Relatively small, but is a matter of life or death**

Life Expectancy

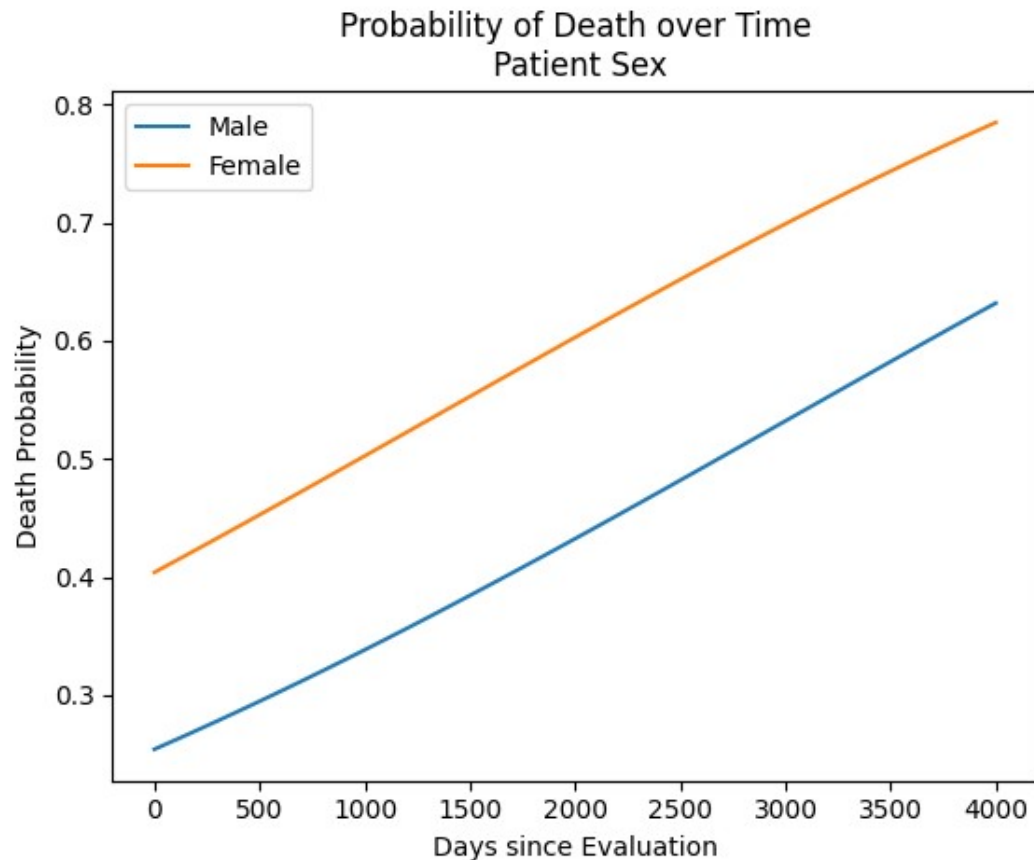
Disease Stage



- **Stage 1 has the lowest death rate**
- **Stage 2 is about 5% above stage 1**
- **Stage 3 is another 5% with Stage 4 another percent more**

Life Expectancy

Sex



- **Woman Have a much higher risk of dying with about a 15-20% difference over time**

Conclusion

- Predict Alive vs Dead
- Visualize rates over time
- Evaluate various feature importance

Thanks for Listening